

Final Project Modeling NAVSTAR System

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Purpose

The purpose of this project was to successfully model the 24 essential components of the GPS (Global Positioning System) satellites placed into orbit by the Department of Defense over the last handful of decades. The history of the modern GPS satellites have their roots embedded deep in the Space Race of the 1960's. When the USSR launched Sputnik and had it transmitting a signal, scientists realized that they could determine the position of the satellite using the Doppler effect, subsequently they determined that the position of a receiver on the ground could be determined by the distance from a satellite. The first line of satellites in the GPS family was Transit which was created for the military purpose of tracking the Navy's submarines and creating more accurate maps of the Earth's surface. The DoD replaced Transit in 1996 with the modern GPS satellites that were opened up to civilian use [a].

Mission Goals and Objectives/Results and Analysis of Orbit

The ultimate goal of the GPS satellites is to provide continuous global navigation and timing services to the public at any point in the world, especially the United States as the satellites were created by the DoD. What this means specifically in STK is that no matter where you are on the globe, you should always have a connection to a handful of satellites at any

moment. In reality there are actually 31 GPS satellites orbiting the Earth but only 24 are technically required for full coverage [b]. In my representation I only chose to model the 24 required as pictured below in Figure 1.

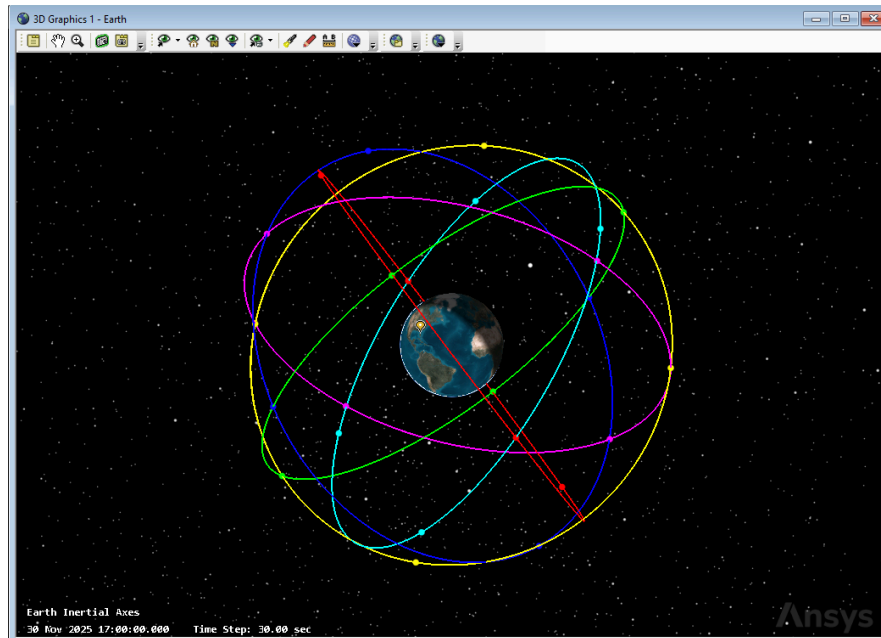


Figure 1: 24 required GPS satellites in MEO

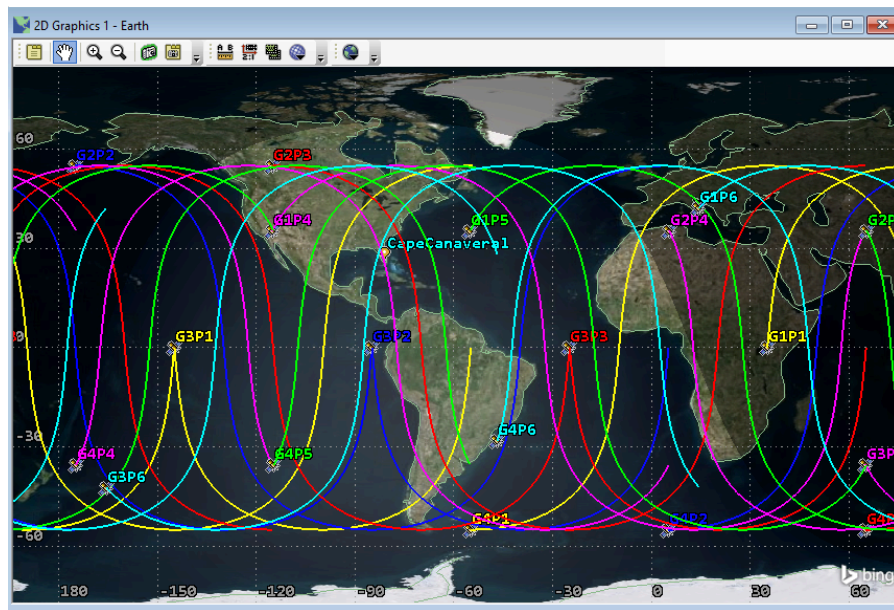


Figure 2: Ground track of 24 satellites

As for the specifications of these satellites, they are all in MEO. Specifically they lie on 6 unique planes all with inclination angles of 55 degrees. In my model there are 4 satellites per plane with true anomaly spacing of 90 degrees. It is important to note that in reality the satellites are likely not in perfect circular orbit so a better choice might have been to space them according to the mean anomaly. They all have altitudes of 20,189 km and eccentricities of 0. The only difference between these satellites is actually their RAAN. Since there are six unique planes, each plane is separated from the last by a RAAN of 60 degrees. While Orbit Wizard allowed for ease of setting these angles, it is important to note how RAAN is determined through launch, which will be covered later in the report.

In order to fully explore how the GPS system works I decided to use a variety of STK tools including access periods, sensors, and coverage analyses. Starting with access periods I found that with all 24 satellites orbiting there are numerous connections both between satellites and ground locations at any given moment visualized below.

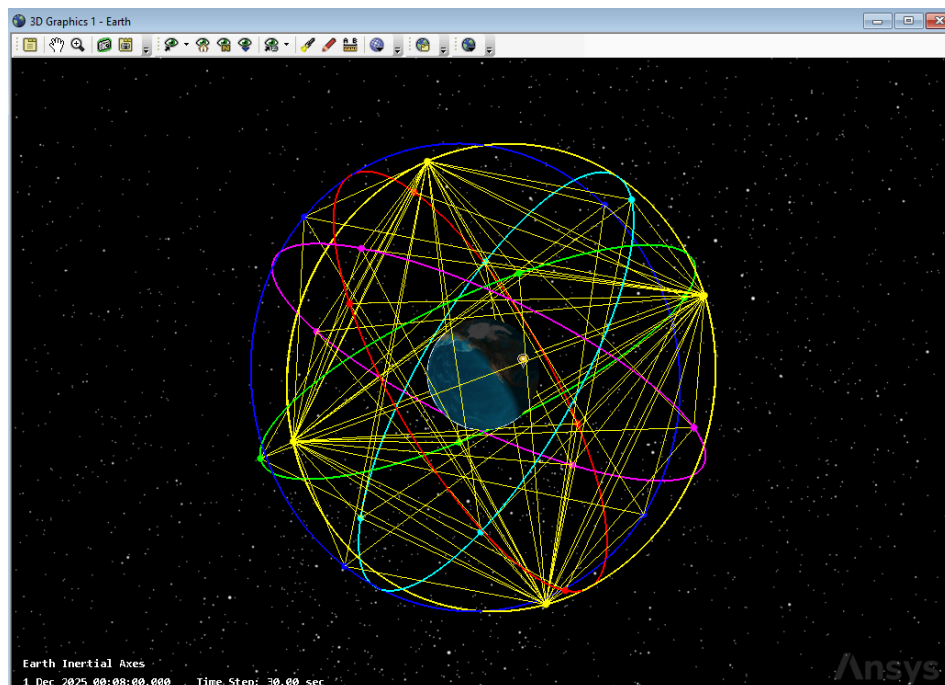


Figure 3: Accesses from 4 satellites to all others

In Figure 3 I only calculated the possible accesses from 4 satellites in one plane with all other available satellites, as we can see there are plenty of connections among satellites as well as between satellites and any locations on the ground.

Next I decided to dive deeper into exactly how the amount of available satellites and their respective positions would affect coverage both globally and within specific countries. I began by defining a coverage area in STK and enclosing just the United States. I then used ground coverage analysis coupled with a figure of merit (FOM) which would allow me to map how many satellites are available in the United States at a given time. It is important to note that for my FOM I opted to define it as the minimum number of assets in view, which essentially tells me at that given moment the least amount of satellites available for coverage. I first mapped the entire thing with all 24 satellites as assets (in use) and produced the following map. Please note for Figures ?-? that the bright green coloring indicates that multiple satellites are available, with the lighter shades corresponding to fewer and fewer satellites being available

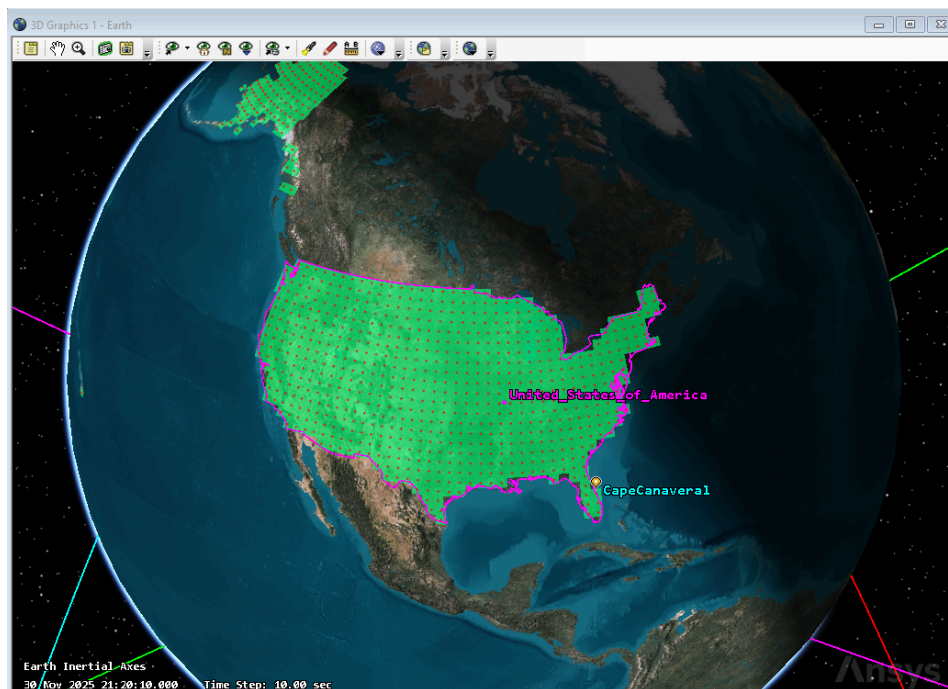


Figure 4: Ground coverage with FOM using 24 satellites for the USA

As we can see with all 24 satellites available the coverage across all of the United States is very strong, with all locations having access to multiple satellites. After determining that the 24 satellites indeed provided strong coverage over the entire region I decided to see the coverage that one satellite may provide.

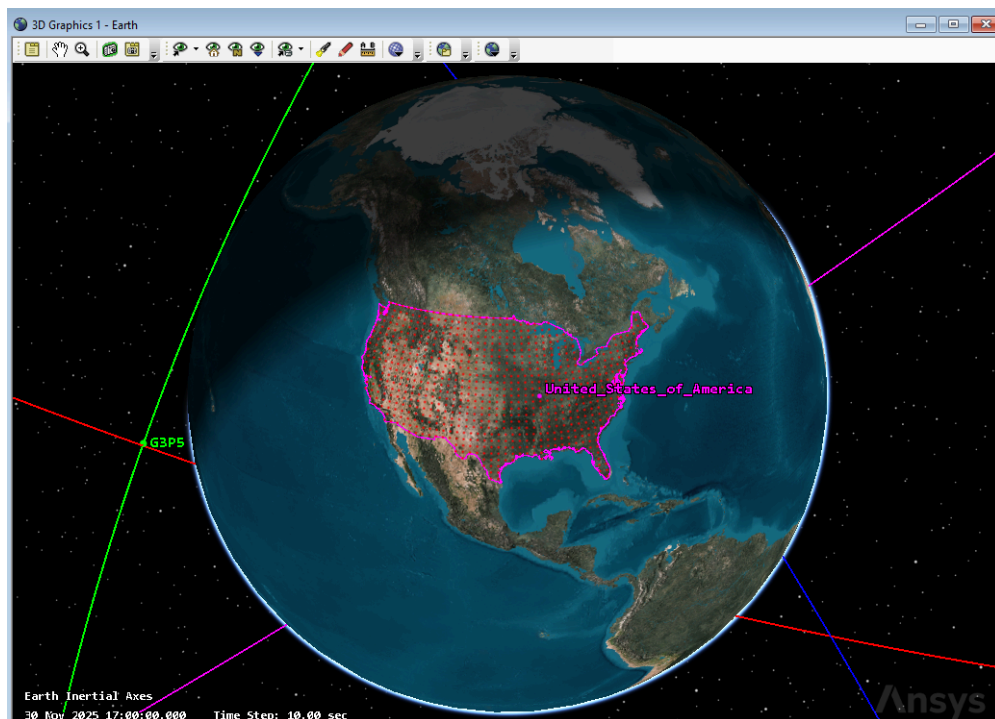


Figure 5: Ground coverage with FOM using 1 satellite for the USA

As we can see with only one satellite active no region of the United States has any kind of access, while this is at one specific moment in time, if I had shown the availability over a time period we would likely see some sort of coverage for a time period. Next I decided to test what coverage might look like if a single plane of satellites were active. I decided to use the plane of satellites with a RAAN of 0 degrees and produced the following.

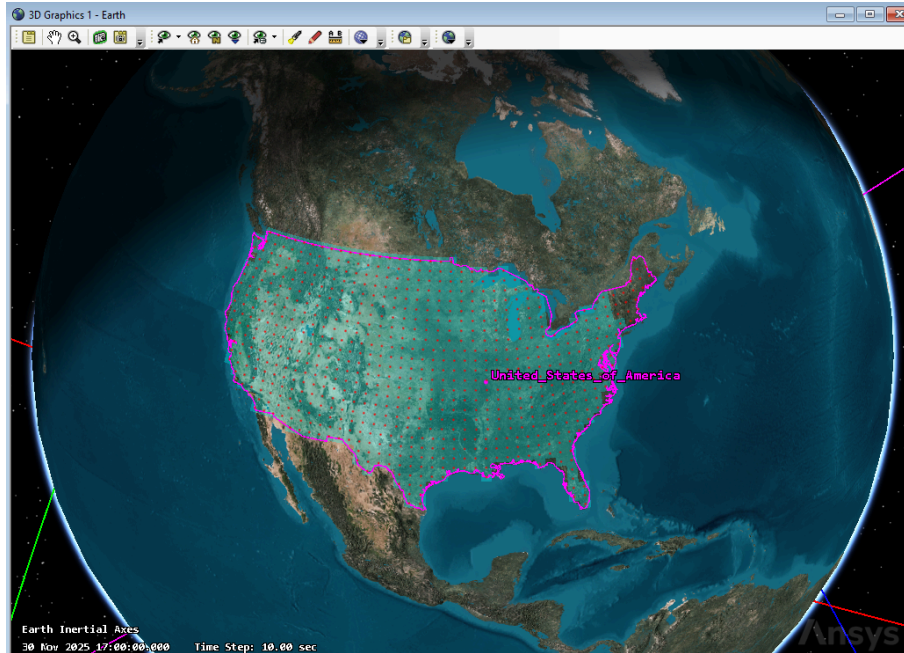


Figure 6: Ground coverage with FOM using 4 satellites (in one orbital plane) for the USA

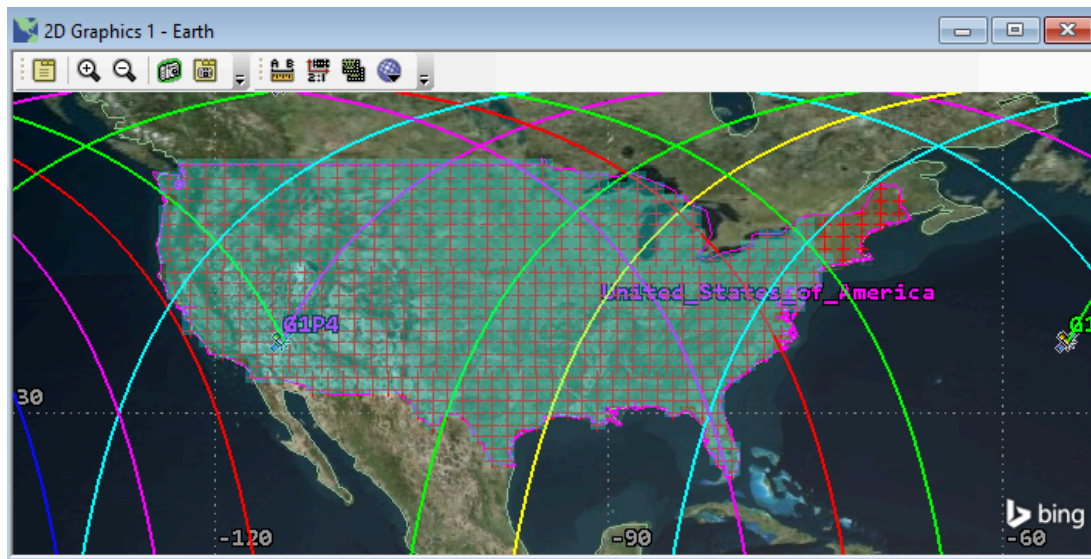


Figure 7: 2D view of ground coverage with FOM using 4 satellites for the USA

As we can see in both the 3D and ground tracks that the single plane of satellites provides some coverage for a majority of the United States, but really only has access to 2-3 satellites at a time, hence the lighter shading on the maps. On top of the low coverage, we see that the northeast region of the United States is actually completely out of coverage, at no point would

this region be available to one of the satellites in the orbital plane chosen. Next I took the satellites in that same plane, but only activated 2 of the satellites as opposed to all 4, namely the ones defined with starting true anomalies of 0 and 90 degrees.

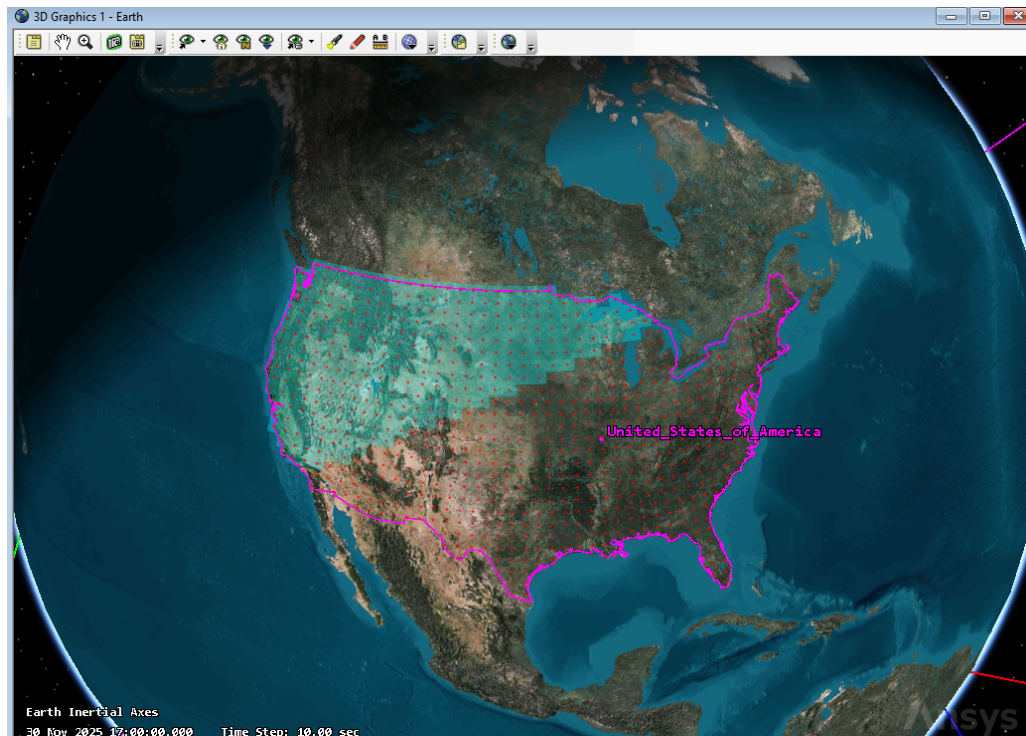


Figure 8: Ground coverage with FOM using 2 satellites (in one orbital plane) for the USA

As we can see roughly half of the continental United States has some kind of coverage and the portion that does would only have access to 1 or 2 satellites at most.

These different scenarios allow us to more clearly understand the importance of the amount of satellites in the GPS orbit. While this is only for the continental United States it is important to note that it may be possible for fewer satellites to be active and still provide great coverage, but the end goal of GPS is complete and fast coverage over the entire world. I then chose to map similar scenarios as above in different regions, namely Australia and the entire world.

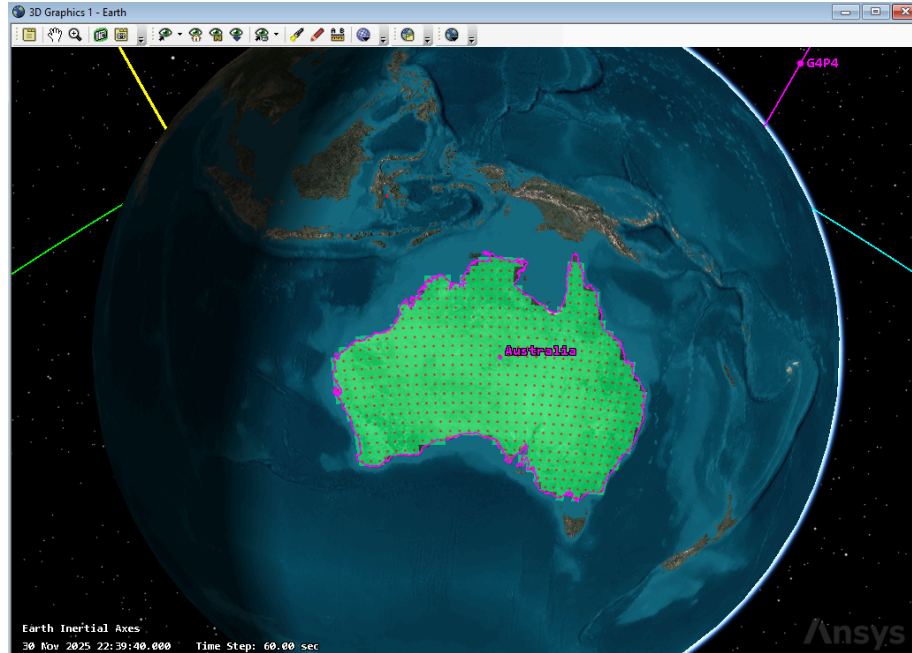


Figure 9: Ground coverage with FOM using 24 satellites for Australia

Figure 9 shows us similar results to the ones found when we mapped the United States, showing us complete coverage everywhere. Next I reduced the assets to just a single plane, the same as before, with a RAAN of 0 degrees.

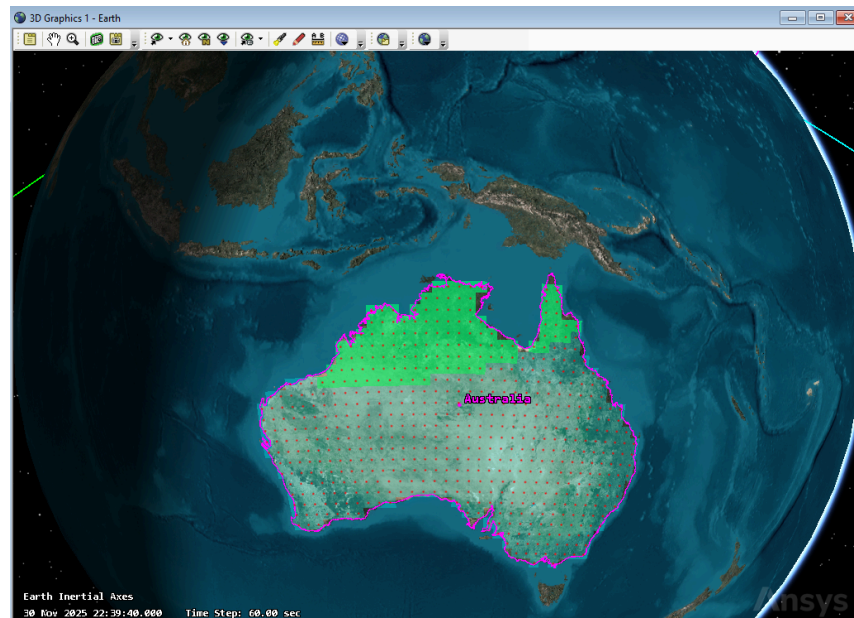


Figure 10: Ground coverage with FOM using 4 satellites (in one orbital plane) for Australia

In Figure 10 we actually see a difference between coverage in Australia and the United States despite having the exact same assets. In Australia we see the northwest region is not actually affected by the reductions of active satellites, maintaining strong coverage. However the remainder of Australia has lost some coverage only having access to 1 or 2 satellites at some time. This comparison shows us the importance of having multiple satellites in each orbit and multiple orbits by visualizing the coverage differences by simply changing where you are on the Earth. Finally I decided to analyze similar scenarios as before but this time with the entire globe as the coverage area.

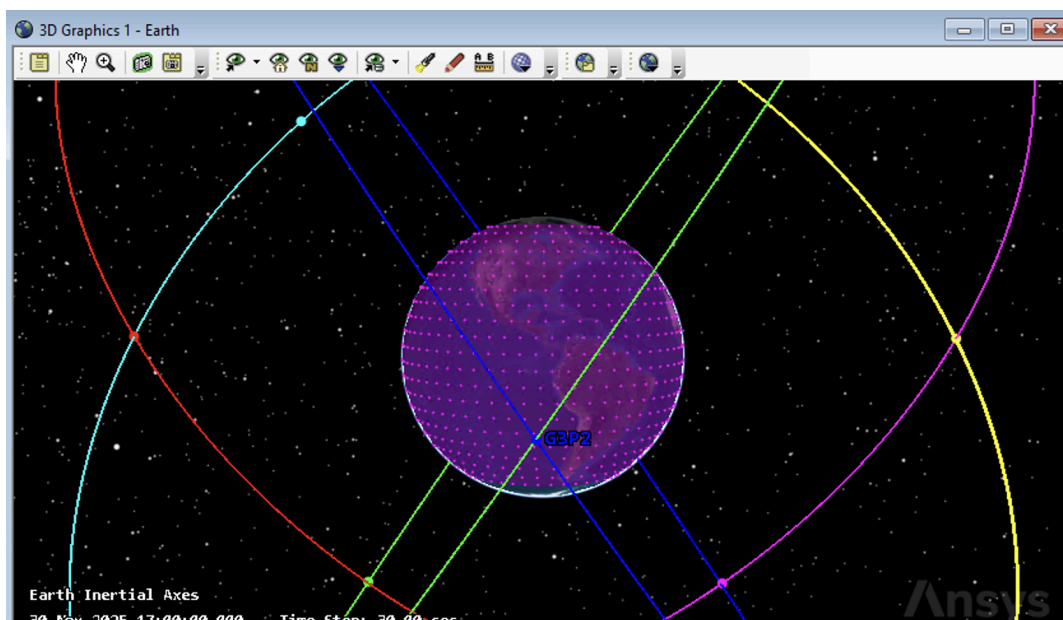


Figure 11: Ground coverage with FOM using 24 satellites for the world

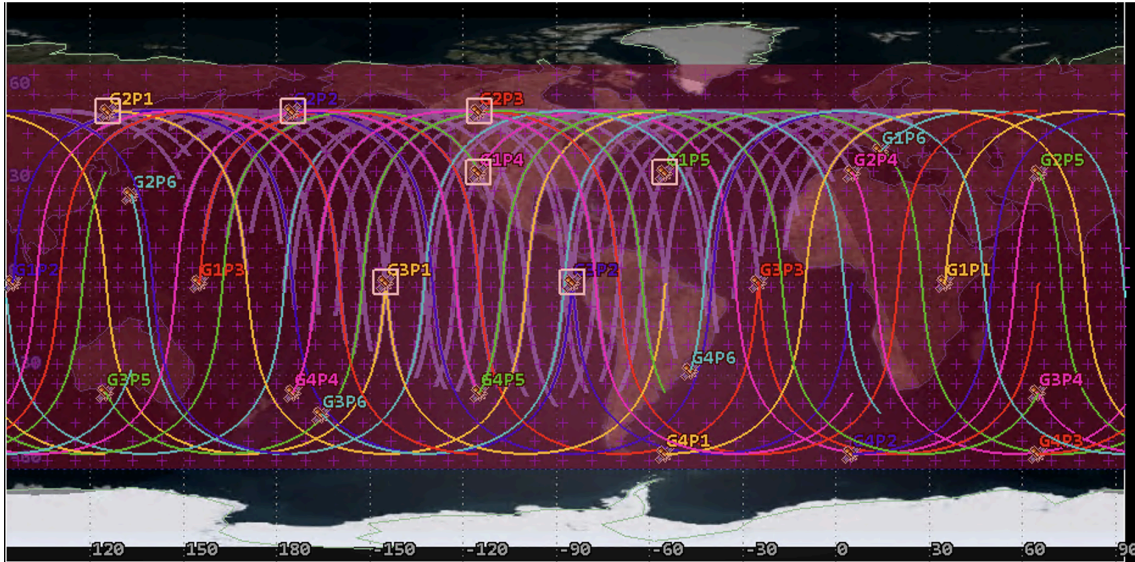


Figure 12: Ground track coverage with FOM using 24 satellites for the world

As we can see we get great coverage at every point of the Earth. During this portion of the modelling I decided to define the area of coverage such that the poles were excluded as the mission is to provide positioning to the public and people do not live in the poles. Next I again explored how coverage would be affected by only having a single plane activated (RAAN = 0 at start).

As we can see there is a lot of discrepancy in coverage depending on the location on Earth, with areas like the Atlantic Ocean and northeast USA experiencing no connection. However some areas remain largely unchanged in terms of coverage, with countries closer to the equator maintaining the same level of coverage and the USA having reduced but available coverage. Finally I modelled the coverage of the world with only two of these satellites active (starting true anomalies of 0 and 90 degrees).

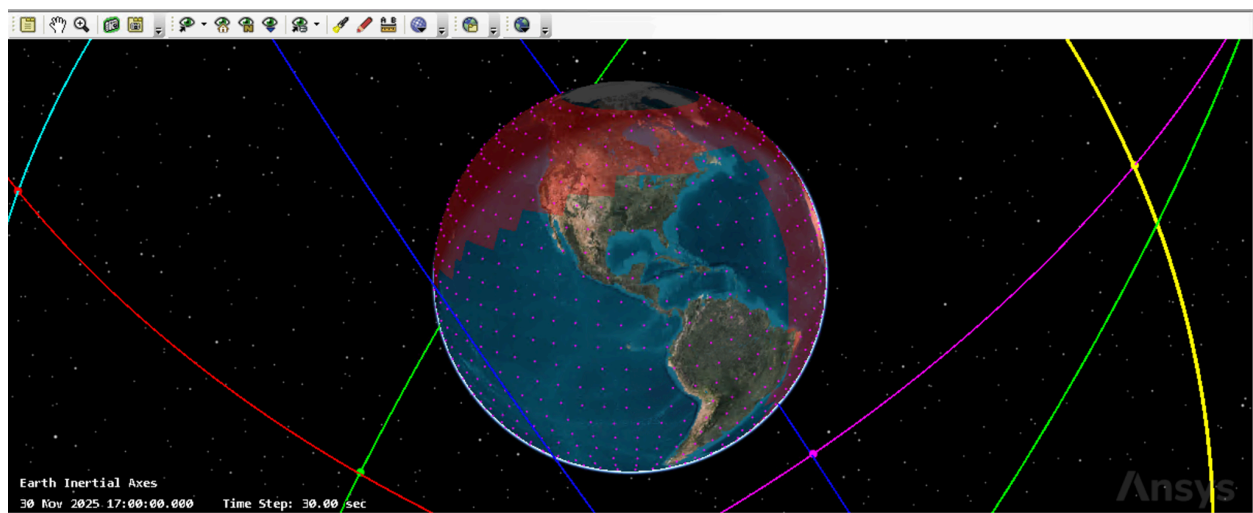
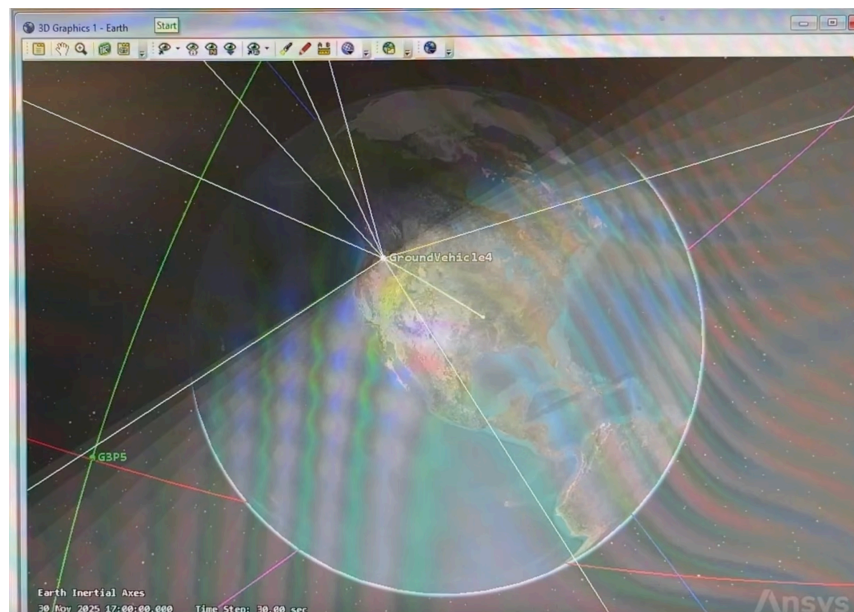


Figure 14: Ground coverage with FOM using 2 satellites (in one plane) for the world

As expected we see the same trends as when we modeled this scenario in the USA and Australia. A large majority of the world loses any type of connection and the places that do retain connection are reduced to only 1 or 2 satellites at most.

For the next portion of this section I decided to explore how sensors worked on moving ground objects. In order to do this I created a car that would start in the northwestern United States and make its way to the southeast. I added a sensor on top of the car that would allow me to model the different accesses that the car would have along its journey. It is important to note the specifications of the implementation of the sensor. When creating a sensor you can set the angle of the conic section from anywhere between 0 degrees (straight up) and 90 degrees (a flat

plane). A couple factors that went into my deciding of what angle to use was what is typically used for receivers and accounting for any noise from surrounding buildings, other signals, etc. I decided to use an angle of 13 degrees from the ground in order to remove noise, as this is typically around the angle most GPS receivers operate [c]. Please find the video below.



Video 1: Availability of satellites to moving car

As we can see in the video at any given moment along the car's path it has multiple satellites available to it. This really encompasses the goal of our mission which was to ensure timely and accurate location data to anyone on the Earth.

Discussion of Goals and Orbit

As visualized in the section above we can see that I was able to model the 24 satellites required for positioning and timing information. I used a variety of tools in STK including orbit wizard, coverage definitions, figures of merit, ground vehicles, sensors, and access periods. While the end goal of the GPS satellites has been produced it is also important to dive into how

these satellites came to be in their orbits, from the launch to any transfer maneuvers performed to place them in MEO. If I had more time to put into this portion of the mission I would have attempted to better model the satellites orbits, I took their eccentricities to be 0 when in reality they would have some slight elliptical shape to them. Another thing I would have begun to explore would have been computing the access periods between multiple ground locations or objects and attempting to determine if there was any combination of locations that would have produced no connection to a satellite.

Modelling of Launch and MEO Transfer

While researching exactly how these satellites were launched, I learned that each of the 31 currently orbiting the Earth, they were all launched into a LEO parking orbit by a variety of rockets including but not limited to the Delta 4, Atlas 5, and SpaceX's Falcon 9. While specifics of launch characteristics were difficult to determine based on public information I did my best to model the launch and maneuvers as they happened. I began by modelling the launch using the basic characteristics granted by astrogator in STK. For the actual launch I had to determine the Az./Ele. angles to launch the satellite given they were all launched at Cape Canaveral by the Department of Defense and the US Air Force. During my research I learned that the rockets actually only took the satellites into a sort of parking orbit where the rockets would then return to Earth, but the satellites themselves would use an on board propulsion system to perform a transfer to their final orbit [d]. After being launched into a LEO parking orbit they would burn into an elliptical transfer orbit until an insertion burn once they reached the desired apogee altitude. I decided not to include the landing or crashing of the first stage of the rockets since the main purpose of this mission was to model the satellites and not the rockets themselves. Also due

to computational difficulties, it seemed the computer was having trouble computing all 24 launches so I opted to launch 2 and expand upon how you would go about launching the other 22.

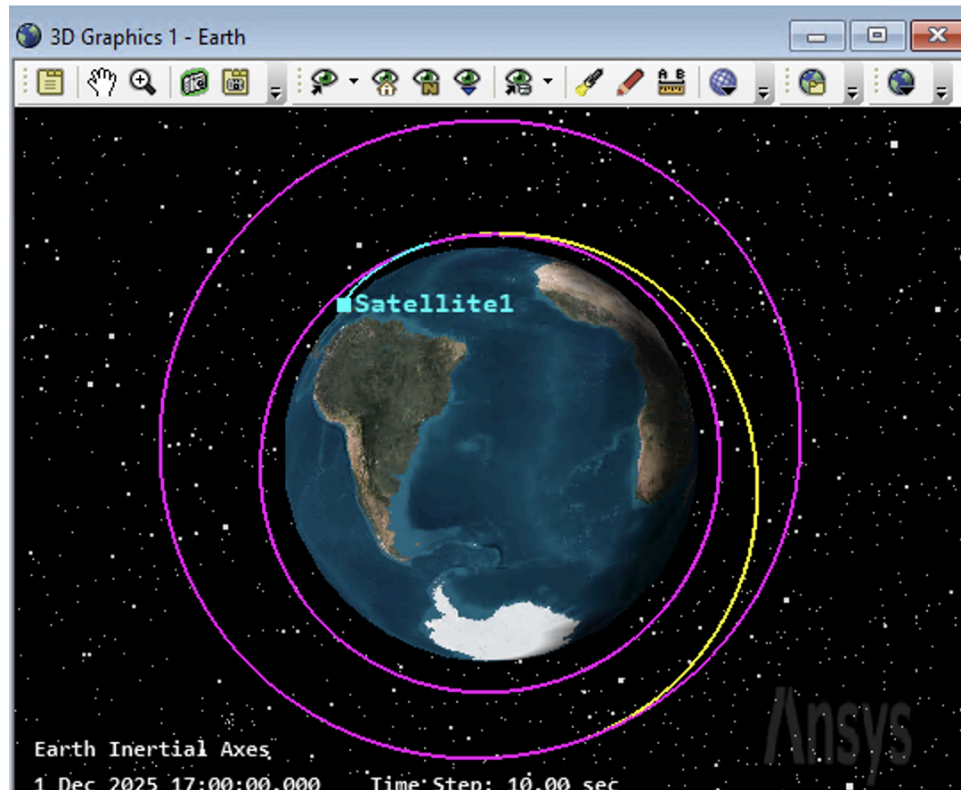


Figure 15: Outline of mission from launch to MEO

As pictured in Figure 15 we can see the mission from launch to its parking orbit into the Hohmann transfer and finally the insertion into MEO. While we did not know the required delta v for this entire mission I decided to calculate what would be required based on the default settings for dry mass and fuel capabilities since I could not find accurate information on a lot of the rockets. However before this I needed to determine the Azimuth angle to set in the launch in order to achieve an inclination of 55 degrees. Using the geometric relationship between azimuth, inclination, and latitude we find the following equation.

$$Az = \arccos\left(\frac{\cos(i)}{\cos(\phi)}\right)$$

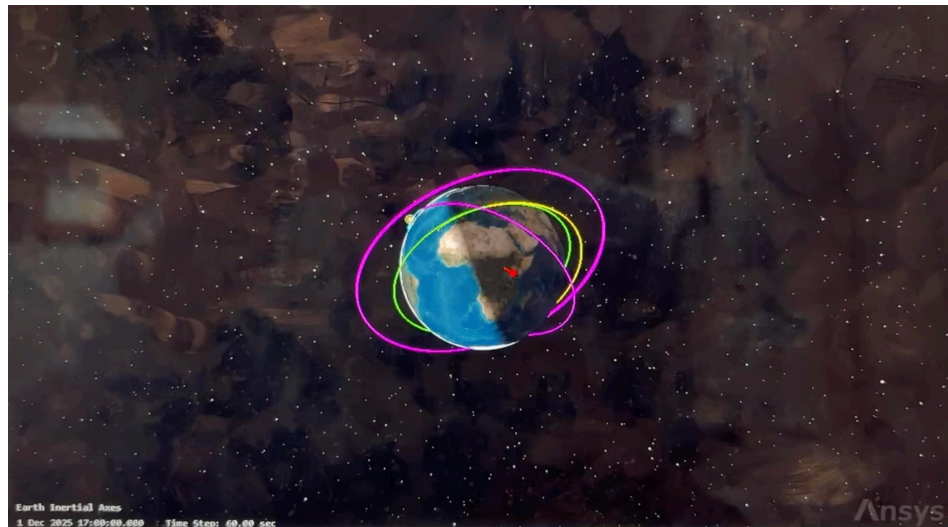
Equation 1

Understanding ‘i’ is inclination and ‘φ’ is latitude we plug in the values 55 degrees and 28.5 degrees for an Azimuth of 49.3 degrees. Please find the calculated velocities for the launch below.

Vx:	7.1694673756618794	km/sec
Vy:	2.7684061358248182	km/sec
Vz:	2.3115892887734240	km/sec

Figure 16

After the launch the satellite would orbit once before beginning the Hohmann transfer into MEO which produced a required first delta v magnitude of 442 m/s for beginning the transfer orbit and a second to begin circular orbit of magnitude 666 m/s, for a total delta v of 1108 m/s. Please find the video of the launch of two satellites below.



Video 2

As a discussion of what’s going on in Video 2, we should note that of our 6 planes of orbit, the only difference between them is their RAAN, which is largely controlled by the timing

of the launch. Given that the Earth completes a full orbit in a day, if we wanted to separate the RAAN's by 60 degrees then we should launch them roughly 4 hours apart. This can be seen in the video, as the first satellite launches around 17:00:00 on Dec 1 and the second around 21:00:00 on Dec 1. The applications of this going forward would be as such, once you had a satellite in each orbital plane, you would simply have to wait until the RAAN lined up again before you could launch another into an existing orbit and attempt to space them roughly 90 degrees apart to avoid collision and maximize efficiency in spacing.

Discussion of Launch and Maneuvers

While I was able to successfully model the launch of the satellites as well as their transition from parking orbits into their final MEO orbits. However there were some likely discrepancies between my model and the actual mission plan that I could expand upon, possibly in part B. For example if I could have found more accurate data on the rockets my launch model might have been a bit more accurate as well as the required delta v 's discussed earlier. Also if I had more time to complete this I would have attempted to create all 24 satellites from launch to orbit as opposed to demonstrating the launch of a couple and then using orbit wizard to solely model their orbit once in place. One of the difficulties I ran into while completing this project was the time frame of orbits and my analysis period, I couldn't seem to get STK to allow a satellite to propagate for an infinite amount of time in astrogator.

Works Cited

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